

Technology Adoption of MSMEs in Food Supply Chain: The Role of AI in Traceability and Transparency

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Abstract— Food consumption is increasing significantly on global stage due to social and economic growth, leading to higher food production and development in food supply chain (FSC). Many firms have been adopting artificial intelligence (AI) in FSC to deal with problems. This research aims to investigate the influence of AI technology adoption of micro, small, and medium enterprise (MSME) FSC in aspect of traceability and transparency in Thailand. As various statistical research discussed in the introduction section points out that nowadays FSC still lack of traceability and transparency, which affects to product consumption of customers. A quantitative approach through an online questionnaire is utilized to gather statistical data, along with conducting the literature review from existing literatures. 92 of MSMEs in Thailand are used as a sample for this research then linear regression is applied for data analysis. The result of the study indicates that AI significantly influences firm's performance in term of traceability and transparency. AI facilitates in sourcing, production, scheduling, and transportation. It provides a better decision making and ensures that all processes in FSC reach standards. However, there are some limitations in technology adoption for MSMEs specifically in micro businesses as they still lack know-how and budget for technological investment.

Index Terms—Food supply chain, artificial intelligence, traceability, transparency, MSME

I. INTRODUCTION

Over 8.1 billion people at present globally, food production and consumption have become way more significant issue comparing to the past. The main goal of the food industry is to provide good foods and other necessities that is suitable for the diversity of human's activities and lifestyle. In order to serve the expanding population, the method of interconnected activities that operate in the food supply chain (FSC) must be transformed significantly [1]. FSC refers to all complex networks between farm and processing activities, distributors, and retailers that operate the concept of 'farm to fork' [2]. The process begins by the raw material until final products are delivered to customers. [2] employed a systematic review on challenges in FSC. The result indicates the three most common issues in FSC, including sustainability, food safety and quality, traceability, and transparency. Among three issues, traceability and transparency gains the lowest percentage of distribution in

term of being mentioned by articles [2], implying that traceability and transparency still do not gain much attention in majority comparing to other two issues, although it is an important topic. FSC with a lack of traceability and transparency will adversely impact the efficiency on FSC [3]. Nowadays food trade becomes globalized, food chain integrity is not concerning only about food safety but also with its origin [4]. In most region, most academic research in firm's performance highlighted more on large firms rather than a small and medium firm [5]. Although MSMEs are a core economic driver in many countries, but still become neglected compared to large firm. Particularly in Thailand, MSMEs play a vital role in the Thai economy contributing to 35.2% of the total gross domestic product (GDP). MSMEs generated GDP value as amounting to 1,784,473 million baht, representing 39.97% of the GDP in manufacturing sector as a whole [6]. These statistical data implies that Thai MSMEs is an important part in Thai economy and should be gained more attention from many departments.

Nowadays traceability and transparency issue, tracking and tracing technology is still not widely standardized. Outdated technologies and traditional manual methods can cause a lack of traceability. [7] had addressed some barriers to AI adoption in FSC. The challenge for AI adoption is due to lacking an appropriate testing system or trial facilities of AI technology, especially in the scope of our study, MSMEs. Local MSMEs still do not have the readiness to become fully invested in technology. Therefore, this research aims to examine the influence of AI technology adoption in FSC on MSMEs' business performance in aspect of traceability and transparency, along with studying the gap that may occur for the adoption.

II. LITERATURE REVIEW

A. Food Supply Chain

FSC is the network of several members who participate in the transformation of raw materials and semi-finished products to final foods delivered to consumers [8]. FSC involves in combination of processes and operations which aim to transform raw products from their original source to final food products consumed by customers [9]. The process of FSC

usually starts with the production of raw agricultural ingredients from farmers in the farm sector [10]. All raw ingredients are well prepared and stored before they undergo manufacturing. All types of ingredients are transformed into edible form as food. Products that reach the standard will be transported and distributed to retail stores. Lastly, customers can finally purchase final products from the retailer. In Thailand, manufacturers keep improving themselves to achieve world food standard and quality as well as social requirements such as domestic customer, oversea customer, and the government. MSMEs in Thailand play an important role mainly in processed fruit and vegetable products. The Thai food industry performed well in term of food export with the rank of 12th the world largest food exporter in 2023 and a growth rate of 3.2% [11].

B. Food Traceability and Transparency

[12] defined food traceability as part of logistic management which captures, stores, and transfers information about food in the entire stages of FSC so that products can be verified to control quality and safety, tracked forwardly, and traced backwardly at any time required. Three main processes are required for a tracking system in FSC, including the transformation of analog data to digital data, transmission of digital data from a local source to the database, lastly the user assess and collaboration [13]. Transparency refers to the disclosure of a product's information, such as production process, ingredients, environmental and social impacts, certification, label that indicates the product's quality and so on [14]. [13] mentioned that the major cause of food contamination is from lack of traceability and transparency. As food takes a long path from producer to customer, food contamination can occur easily at any stage of FSC leading to a negative domino effect. Once it happens, it takes a long time to point out the exact source and reason of contamination and specifically identify the contaminated products. This issue can become a crucial concern to all members in FSC, especially those who are still lacking digital and technology skills [4].

C. Artificial Intelligence

Artificial Intelligence (AI) is a sub-field of computer science involving with creation of innovative machines and computer systems that can perform any task which requires human intelligence [15]. These innovations have the potential to directly impact both the manufacturing processes and the details of several products and services, with important implications for productivity, employment, and competition [16]. [17] employed a systematic review of AI applications in supply chain and classified AI according to their techniques into four sub-fields. First sub-field includes Artificial neural network and Rough set theory. The second sub-field includes Machine learning, Expert systems and Genetic algorithms. The third sub-field includes Fuzzy logic. Lastly, forth sub-field includes Agent-based systems. This study is going to highlight on the outstanding AI techniques based on the result of application in the supply chain and adapting to FSC for this study. According to the result from [17], machine learning is the most dominant type of AI that has been used throughout all processes in supply chain. Genetic

algorithm is the second most used type of AI which is used in five out of six processes in supply chain. Artificial neural network, Expert system, Fuzzy logic, and Agent-based system are applied in three processes. However, only ANN is engaged in purchasing and procurement, production, and inventory management which are core processes that directly affect the product's quality and major management in all supply chain. Therefore, this study will highlight three potential AI sub-fields, including machine learning, genetic algorithm, and artificial neural network

a) Machine Learning: Machine learning (ML) is a subset of AI that facilitates the system to be capable of identifying patterns and making decision by learning and improving from experience and previous data set without being programmed [18]. Algorithms are trained using a large collection of two data groups, the training set and the testing set. The algorithm will receive a set of inputs with correct outputs, known as pre-labeled information, to compare its actual output with correct outputs and spot errors [19]. Regression is an example of machine learning that will be applied to this research. Regression is used to determine a relationship between the independent variable and the dependent variable. It instructs a model using a dataset of X and Y and uses the trained model to predict the output of new input data [26].

b) Genetic Algorithm: Genetic algorithm (GA) refers to the principle of natural selection and algorithms inspired by the biological evolution of living beings [20]. The fittest chromosome will survive to generate new offspring, and this process is replicated for multiple generations [21]. The weaker chromosome will be eliminated. This technique is used for optimization problems by finding the maximum or minimum value. [22]. There are five components in GA, which are chromosome representation, selection, crossover, mutation, and fitness function computation [23]. Selection, crossover, and mutation are considered as three core operations in GA. Selection is where a chromosome in the population will be in a structure that hold information related to a potential solution with a combination of weights [24]. Crossover is to combine chromosomes of two parent to produce new offspring that is phenotypically like the parents [25]. Mutation is proformed intentionally to stop falling of all solution in the population to an optimal level of the solved problem [21].

c) Artificial Neural Network: Artificial neural network (ANN) is a processing technique that imitates the working of brain cell and the nervous system of humans. Its main goal is to figure out the pattern in dataset and do the clustering according to natural grouping in the data [26]. The neuron, a special component of an ANN, plays an important role in imitating the human brain to perform tasks. Neurons communicate by transmitting electrical and chemical signal to neighbors through a neuron's branch called an axon, which is further segmented to collaterals [27]. Neurons transfers signals to the receiving end by the formation of connections between synapses, neuromuscular junctions at the end of collaterals, and neighboring neurons. At the receicing end, neuron's dendrites receive signals via synapses and merge them with neuron's body called the soma. Neurons with stronger synaptic

connections will have more weight and greater influence on one another. These massive interconnections of neurons collaborating to provide excellent abilities with the same level as the human brain.

D. Related Research

AI is a highly influential invention that provides a massive impact world in several aspects of human beings. Traceability is one of the major concerns in FSC as it allows producers, consumer, inspector, and government agencies to access the level in food safety issues, and also determining when and where issues may have emerged, leading to the quick elimination of harmed products in FSC [28]. [29] took sample of Italian and foreign extra virgin olive oil to access their actual graphic origins analyzed the data with ANN. The result indicates that 87 out of 92 samples are classified correctly according to their origin, providing 94.6% of classification accuracy.[30] integrates an ANN model for tracking forwardly and tracing diversly with real-time evaluation of food quality along the entire supply chain, along with giving the final determination of food grade in three different levels based on their quality. [31] developed an advanced AI system based on multiple sensors and ML to enable real-time monitoring and detection of the product's freshness. Consequently, this study proposes a major hypothesis with three sub-hypotheses as illustrated in Fig. 1 to study the influence of AI adoption in FSC in terms of traceability and transparency as follows:

- H_1 : AI can significantly enhance traceability and transparency in FSC.
- H_{1a} : ML can significantly enhance traceability and transparency in FSC.
- H_{1b} : GA can significantly enhance traceability and transparency in FSC.
- H_{1c} : ANN can significantly enhance traceability and transparency in FSC.

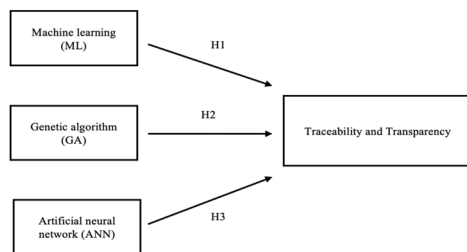


Fig. 1. Theoretical framework illustrating the relationship between AI and traceability and transparency.

III. METHODOLOGY

We conducted a survey as primary data for analyzing all variables along with employing a systematic review of published literature as secondary data. The research adopts quantitative method through the questionnaire survey to approach data collection and analysis. The content of questionnaire is developed based on what level of AI adoption the firm has and key metrics to evaluate the firm's performance. The overall population is in firm level covering micro, small and

medium sized enterprises (MSMEs) food and beverages business in Thailand, particularly in the manufacturing sector. The firms that were eligible to participate for the questionnaire or the overall population were retrieved from a website of relevant department known as the Office of Small and Medium Enterprise Promotion (OSMEP) which is a Thai government department responsible for supporting SME firm in Thailand [32]. Another source is the database from Food Intelligence Center (FIC). FIC is operated under the Thailand's ministry of industry whose main objectives are to research, collect, and follow news in Thai food industry [33]. The key respondent is normally the business owner, manager or employee who are appropriate to fill the questionnaire referring to their knowledge and expertise in FSC's performance of their firm [34]. The questionnaire developed in this study comprises of four sections and was conducted by a Likert scale from 1 – 5. Sections one and two ask about the demographic information of participants and the information of the firm respectively. The third section measures the level of AI adoption in FSC based on various research [17, 32, 35-37]. Lastly, section four investigates the level of performance in traceability and transparency of a firm's FSC based on several studies [28, 38-43]. The questionnaire is organized in online format as a Google Form and handed out to respondents via email.

According to the database of OSMEP and FIC, there are 1,412 and 7,521 firms respectively which provide the population size of 8,933 firms in total. Krejcie and Morgan's equation is applied to determine the sample size for the research which determined the sample size for this study as 368 firms [44]. In practical the researcher was able to send questionnaire to 280 firms due to limitation of time constrain so, the sample size using for this analysis is 280. However, due to some several undesirable reasons, a number of firms refused to provide internal business information and decided not to complete the questionnaire, consequently 92 of MSMEs were willing to answer the questionnaire. Finally, all questionnaire responses are gathered and employed in multiple regression analysis to study the relationship of technology and firm's performance.

IV. RESULTS AND DISCUSSIONS

All 92 samples of the research are from all regions over Thailand consisting of Central, Northern, Southern, Eastern, Western and Northern east. The majority of respondents are from micro firms and in the role of business's owner and employee. Top three categories of core business operations from responses are processed fruit and vegetable production, general food production, and beverages production. The sample includes 49 micro, 22 small, 11 medium, and 10 large-sized firms. The common level of AI adoption in the three sub-field is in a neutral level followed by not adopted and adopted level. From the survey, major reason of AI implementation barriers for MSMEs are financial constraints of the owner as their business scale is not large. Many of them operated business in countryside where technology is still not generalized. Highly adopted level is chosen the most in ANN adoption. ML is most used in calculating a proper amount of production along with selecting a supplier. GA is used by MSMEs in term of

calculating for optimal value including inventory management and demand forecasting. Lastly, ANN is most used for dealing with large database and handling with external factor that may affect to business. The result from running regression indicates that all four hypotheses are supported as illustrated in Table 1.

TABLE I. THE P-VALUE TABLE OF EACH HYPOTHESIS

<i>Hypotheses</i>	<i>p-value</i>	<i>Result</i>
H1: AI can significantly enhance traceability and transparency	0.000	Supported
H1a: ML can significantly enhance traceability and transparency	0.000	Supported
H1b: GA can significantly enhance traceability and transparency	0.000	Supported
H1c: ANN can significantly enhance traceability and transparency	0.000	Supported

To summarized, AI as in the sub-field of ML, GA and ANN have a positive impact toward traceability and transparency in FSC. Several firms adopt AI for their operation to improve processes along the supply chain and deal with management. According to the results of the research, ANN influences traceability and transparency in FSC of MSMEs the most with the standardized coefficients of 0.551, followed by GA and ML with standardized coefficients of 0.513 and 0.479 respectively. ANN distributes significantly in firms' performance. Most firms implement ANN to deal with large set of data for example supplier selection. ANN is also applied by firms for dealing with external situations that affect the business which is market fluctuation or changes from customer's demand. It considers actual historical data of business to figure out a pattern and uses for prediction. GA is applied in term of determining optimal value for processes in supply chain. GA involves in inventory management to calculate the proper quantity and time schedule for production based on weighting between certain and required amount of product. Lastly, they use ML for detection then do processing for real-time tracking to display quantity and condition of products. In term of traceability and transparency performance, most firms can track and monitor the status of products along the entire production. According to limitation of local MSMEs, political policy may enhance technology accessibility. For example, government launch a competition for MSMEs product and the winner could win a prize of getting a support for technological investment and could be an idea for other MSMEs owner as well.

V. CONCLUSION

In conclusion, several firms use AI techniques for their business management in various stages of FSC such as purchasing, production, inventory management, demand forecasting and so on. The result of questionnaire done by respondents, a representative of each MSME from all regions in Thailand, shows that AI adoption has influence toward firm's performance in aspect of traceability and transparency. Every AI-sub field, ML, GA, and ANN are significantly influences firm's performance positively. Responses from questionnaire

convey that the implementation of these AI sub-field can improve the quality of business operations in several ways. They can be used to manage and analyze large amount of data that can be difficult to handle manually. Moreover, AI enhances deciding based on previous data. It used previous data to predict or estimate new value which is helpful for firm to do the appropriate forecasting. Most of the level of AI adoption in FSC is in medium level. Some group of firms, especially micro-sized and small businesses, AI technology is beyond their scope of business. Several respondents mentioned that they are capable of dealing and managing all processes which has low complexity due to their business scale without a requirement of high technology adoption. In addition, AI technology adoption requires high cost of implementation, they may not be able to afford and access the use of technology. High operating cost and lack of control on innovation costs are two of limitation in MSMEs' product innovation [45]. Their return is not worth the investment cost. Consequently, this is a research gap for technology adoption in MSMEs business that can be developed further so that AI technology can be easier to access for various type of business in the future.

REFERENCES

- [1] P. Haessner, J. Haessner, and M. McMurtrey, "Trends and challenges in the food supply chain," *Journal of Strategic Innovation and Sustainability*, 1st ed., vol. 19, pp. 115-124, 2024.
- [2] K. Gurralla, and H. Moncer, "Key food supply chain challenges: A Review of the Literature and Research Gaps," *Operations and Supply Chain Management*, 4th ed., vol. 15, 2022.
- [3] M. Dora, A. Kumar, S.K. Mangla, A. Pant, and M.M. Kamal, "Critical success factors influencing artificial intelligence adoption in food supply chains," *International Journal of Production Research*, 14th ed., vol. 60, pp. 4621-4640, 2021.
- [4] S. Kumperscak, M. Medved, M. Terlav, and A. Wrzalik, "Traceability systems and technologies for better food supply chain management," *Quality Production Improvement*, 1st ed., vol. 1, pp. 567-574, 2019.
- [5] V. Verma, P. Bisht, and S. Joshi, "Sustainable supply chain systems of food and beverages SMEs in Countries: Theoretical perspective," *Journal of Decision Analytics and Intelligence Computing*, 1st ed., vol. 2, 2022.
- [6] No author, "MSME Report in 2/2024," 2024, The Thai Chamber of Commerce. [Online]. Available: <https://www.thaichamber.org/news/view/87/3034/msme-outlook-q2-2024>
- [7] N. Ghag, H. Sonar, S. Jagtap, and H. Trollman, "Unlocking AI's potential in the food supply chain: A novel approach to overcoming barriers," *Journal of Agriculture and Food Research* 18, 2024.
- [8] R. Zhong, X. Xu, and L. Wang, "Food supply chain management: systems, implementations, and future research," *Industrial Management & Data Systems*, 9th ed., vol. 117, pp. 2085-2114, 2017.
- [9] U. Ziemińczyk, and A. Krakowiak-Bal, "Factors influencing food supply chains in the context of exiting food quality schemes," *Polish Academy of Sciences*, 1st ed., vol. 5, pp. 1111-1112, 2018.
- [10] N. Georgiev, "What Is the Food Supply Chain? | 5 Food Supply Chain Stages," BlueCart, n.d. [Online]. Available:

- [11] No author, "Thailand is in 12th world ranking of exporters," Thansettakij, n.d. [Online]. Available: <https://www.thansettakij.com/business/trade-agriculture/584974>
- [12] T. Bosana, and G. Gebresenbet, "Food traceability as an integral part of logistics management in food and agricultural supply chain," *Food Control* 33, pp. 32-48, 2013.
- [13] L. Dong, P.P. Jiang, and F. Xu, "Impact of traceability technology adoption in food supply chain networks," *Management Sciences*, 2022.
- [14] C. Hubbell, L. Monaco, and J. Lusk, "The future of traceability and transparency in the food System,". 2023.
- [15] E. Glover, "Artificial intelligence definition," Built-in, Jul. 2025. [Online]. Available: <https://builtin.com/artificial-intelligence>
- [16] I.M. Cockburn, R. Henderson, S. Stern, "The impact of artificial intelligence on innovation,". National Bureau of Economic Research, pp. 115-146, 2019.
- [17] M. Atwani, M. Hlyal, J. Elalami, "A Review of Artificial Intelligence applications in Supply Chain,". IMT Web of Conferences, st ed., vol 46, 2022.
- [18] A. Chahal, and P. Gulia, "Machine learning in deep learning, International Journal of Innovative Technology and Exploring Engineering,". International Journal of Innovative Technology and Exploring Engineering (IJITEE), 12th ed. , vol. 8, pp. 4910-4914, 2019.
- [19] S.H. Shetty, S. Shetty, C. Singh, and A. Rao, "Supervised machine learning: algorithms and applications,". Fundamentals and Methods of Machine and Deep Learning, pp. 1-16, 2022.
- [20] H. Lingaraj, "A study on genetic algorithm and application,". International Journal of Computer Sciences and Engineering, 10th ed., vol. 4, pp. 139-143, 2016.
- [21] A. Lambora, K. Gupta, and K. Chopra, "Genetic algorithm – A literature review,". International Conference on Machine Learning, Big Data, Cloud and Parallel Computing (COMITCon), pp. 380-384, 2019.
- [22] T. Alam, S. Qamar, A. Dixit, and M. Benaida, "Genetic algorithm: reviews, implementation, and applications,". International Journal of Engineering Pedagogy (IJEP), 2020.
- [23] S. Katoch, S.S. Chauhan, and V. Kumar, "A review on genetic algorithm: past, present, and future,". Multimed Tools Appl, 5th ed., vol. 80, pp. 8091-8126, 2021.
- [24] D. Yun, "Merging multiple independently trained neural networks Based on Genetic Algorithm,". 2024.
- [25] K. Pretorius, and N. Pillay, "Neural network crossover in genetic algorithms using genetic programming,". Genetic Programming and Evolvable Machines, 2024.
- [26] S. Naeem, S. Anam, S. Ahmed, and A. Ali, "An unsupervised machine learning algorithms: comprehensive review,". International Journal of Computing and Digital Systems, 1st ed., vol. 13, pp. 911-921, 2023.
- [27] A. Vasileiadis, E. Alexandrou, L. Paschalidou, and M. Chrysanthou, "Artificial neural network and its applications,". 2024.
- [28] S. Gbashi, and P.B. Njobeh, "Enhancing food integrity thorough artificial intelligence and machine learning: A Comprehensive Review,". A Comprehensive Review. Applied Sciences, 8th ed., vol. 14, pp. 3421, 2024.
- [29] Violino, S., Ortenzi, L., Antonucci, F., Pallottino, F., "An Artificial Intelligence Approach for Italian EVOO Origin Traceability through an Open Source IoT Spectrometer,". Foods, 9th ed., vol. 6, pp. 834, 2020.
- [30] J. Wang, H. Yue, and Z. Zhou, "An improved traceability system for food quality assurance and evaluation based on fuzzy classification on neural network,". *Food Control* 79, pp. 363-370, 2017.
- [31] M.J.I. Tutul, M. Alam, M.A.H. Wadud, "Smart food monitoring system based on IoT and machine learning,". 2023 International Conference on Next-Generation Computing, IoT and Machine Learning (NCIM), pp. 1-6, 2023.
- [32] No author, Office of Small and Medium Enterprise Promotion, 2025. [Online]. Available: <https://thaismegp.sme.go.th/smes>
- [33] No author, Food Intelligence Center, 2025. [Online]. Available: https://fic.nfi.or.th/food-directory-list.php?company_name=&product_cate_id=19&province=&budget=
- [34] M. Bourlakis, G. Maglaras, E. Aktas, D. Gallear, and C. Fotopoulos, "Firm size and sustainable performance in food supply chains: Insights from Greek SMEs,". *Production Economics* 152, pp. 112-130, 2014.
- [35] I. Santoso, M. Purnomo, A.A. Sulianto, and A. Choirun, "Machine learning application for sustainable agri-food supply chain performance: a review,". IOP Conference Series Earth and Environmental Science, 1st ed., vol. 924, 2021.
- [36] H. Anwar, T. Anwar , and G. Mahood, "Nourishing the future: AI-driven optimization of farm-to-consumer food supply chain for enhanced business performance history,". *Innovative Computing Review*, 3rd ed., vol. 2., 2023, pp. 15-29.
- [37] V.L. Bravo, and F. Caniato, "Sustainability performance measurement in the food supply chain: Trade-offs, institutional pressures, and contextual factors,". *European Management Journal*, 4th ed., vol. 42, pp. 633-646, 2024.
- [38] V. Dedeoglu, S. Malik, G. Ramachandran, S. Pal, and R. Jurdak, "Chapter nine – blockchain meets edge-AI for the food supply chain traceability and provenance,". *Chemistry* 101, pp. 251-275, 2023.
- [39] V. Baciuliene, Y. Bilan, V. Navickas, and C. L., "The aspects of artificial intelligence in different phases of the food value and supply chain," *Foods*, 8th ed., vol. 23, pp. 1208 – 1654, 2023.
- [40] M. Krstic, G.P. Agnusdei, S. Tadic, and P.P. Miglietta, "Prioritization of e-traceability drivers in the agri-food supply chains,". *Agricultural and Food Economics*, 42nd ed., vol. 11, 2023.
- [41] R.N. Mefford, "The economic value of a sustainable supply chain,". *Business and Society Review*, 1st ed., vol. 116, pp. 109-143, 2011.
- [42] S. Sudaram, and A. Zeid, "Artificial intelligence-based smart quality inspection for manufacturing,". *Micromachines*, 3rd ed., vol. 14, 2023.
- [43] H. Onyeaka, P. Tamasiga, U.M. Nwauzoma, T. Miri, U.C. Juliet, O. Nwaiwu, and A.A. Akinsemolu, "Using artificial intelligence to tackle food waste and enhance the circular economy: maximizing resource efficiency and minimizing environmental impact: A Review,". *Sustainability*, 2023.
- [44] R. Krejcie, and D. Morgan, "Determining sample size for research activities,". *Educational and Psychological Measurement*, pp. 607-610, 1970.
- [45] P. Chaochotechuang, and S. Mariano, "Alignment of new product development and product innovation strategies: A case study of Thai food and beverage SMEs,". *International Journal of Globalisation and Small Business*, 2nd ed., vol. 8, 2016.